

Cambridge Lower Secondary Checkpoint

MATHEMATICS**0862/02**

Paper 2

October 2024

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

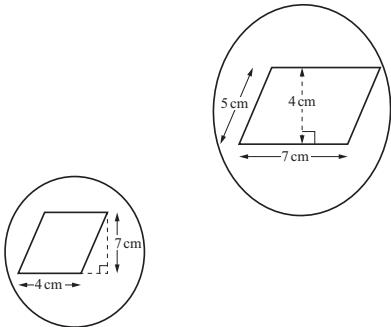
Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **10** pages.

Mark scheme annotations and abbreviations

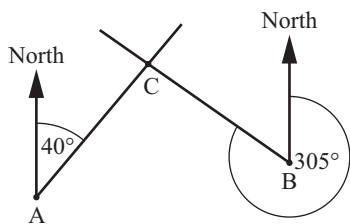
FT	follow through after error
SC	special case mark
cao	correct answer only
dep	dependent
isw	ignore subsequent working
nfww	not from wrong working
oe	or equivalent
soi	seen or implied

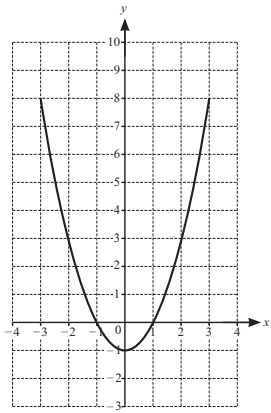
Question	Answer	Marks	Part marks	Guidance
1	125	1		Do not accept 5^3
2	5.8×10^{-2}	1		Accept any clear indication.
3	6	1		
4	135(°)	2	Award 1 mark for $180 - (360 \div 8)$ or $(8 - 2) \times 180[\div 8]$ oe	oe, e.g. 1080
5	4.1 oe	1		oe, e.g. $\frac{41}{10}$
6(a)	Any two different equations of the form $y = 5x + c$ where $c \neq 2$	1		Both equations correct for the mark. Accept lines not in the form $y = mx + c$, but not those equivalent to $y = 5x + 2$ c does not have to be an integer and can be 0
6(b)	-3	1		Accept (0, -3) or 0, -3 Do not accept $y = -3$
7	True ☒ False ☐ ☐ ☒	1		Both answers correct for the mark. Accept any clear indication.

Question	Answer	Marks	Part marks	Guidance
8	106 (cm ³) cao	3	Award 2 marks for 106.3 to 106.4... or $\frac{4232\pi}{125}$ (cm³) or Award 1 mark for $\pi \times 2.3^2 [\times 6.4]$ If 0 or 1 scored, award 1 mark for <i>their</i> more accurate answer shown then rounded correctly to 3sf.	Award 2 marks before any attempts at rounding and if trailing zeros included, e.g. 106.0 Accept values of π between 3.14 and $\frac{22}{7}$ 1 mark may be implied by 16.6... or $\frac{529\pi}{100}$ or 106.2[4]
9		1		Both answers correct for the mark. Accept any clear indication.
10	True False ☒ ☐ ☐ ☒	1		Both answers correct for the mark. Accept any clear indication.

Question	Answer	Marks	Part marks	Guidance
11(a)	78 (kg)	3	Award 2 marks for correct method to find the total, e.g. $4 \times 50 + 6 \times 70 + 8 \times 90 + 2 \times 110$ or Award 1 mark for three correct midpoints from the four midpoints 50, 70, 90 and 110 or for correct method to find an estimated total using consistent points within range.	For 3 marks, isw answer $60 \leq m < 80$ following 78 found. Implied by 1560 or $200 + 420 + 720 + 220$ e.g. using end points of intervals, may be implied by $160 + 360 + 640 + 200$, 1360 or 68 $240 + 480 + 800 + 240$, 1760 or 88
11(b)	A correct reason, e.g. - The data is grouped. - We don't know exact values. - Midpoints are used.	1		Accept equivalent correct reasons.
12	$8x$ and $3y + 2$	2	Award 1 mark for one correct answer.	

Question	Answer	Marks	Part marks	Guidance										
13	$5 < x < 10$ And $n \leq 2$	3	Award 1 mark for $5 < x < 10$ and Award 2 marks for $n \leq 2$ or Award 1 mark for correct first step, e.g. $3n \leq 18 - 12$ or $n + 4 \leq 6$ or better.	2 embedded, answer 2 or $n = 2$ scores 1 mark only. For this 1 mark, accept any inequality sign or equals sign.										
14	<table><tr><td></td><td>Red</td><td>Yellow</td><td>Pink</td><td>Green</td></tr><tr><td>Prob</td><td>(0.25)</td><td>(0.1)</td><td>0.3</td><td>0.35</td></tr></table>		Red	Yellow	Pink	Green	Prob	(0.25)	(0.1)	0.3	0.35	2	Award 1 mark for $P(\text{Pink}) = 0.3$ or for $P(\text{Green}) = 0.35$ or for <i>their</i> $P(\text{Pink}) + \textit{their} P(\text{Green}) = 0.65$	Accept equivalent fractions and percentages.
	Red	Yellow	Pink	Green										
Prob	(0.25)	(0.1)	0.3	0.35										
15	(\$)6859	2	Award 1 mark for $8000 \times \left(1 - \frac{5}{100}\right)^3$ oe or for 7220 seen.	For the 1 mark, ignore if they subtract <i>their</i> answer from 8000 implied by answer 1141 Note 5% is not sufficient for $\frac{5}{100}$ Common incorrect answer 6800 from decreasing by 15% scores 0										
16	<table><tr><td>Input</td><td>Output</td></tr><tr><td></td><td>1</td></tr><tr><td>7</td><td></td></tr><tr><td>9</td><td></td></tr></table>	Input	Output		1	7		9		3	Award 1 mark for each correct answer.	The 7 and -9 can be reversed.		
Input	Output													
	1													
7														
9														

Question	Answer	Marks	Part marks	Guidance
17	32	2	Award 1 mark for $\frac{8}{11+8+3} \times k$ oe where k is an integer ≥ 70 or for $70 \div (11+8+3) [\times 8]$ oe or [total =] 88 soi or for $11k$, $8k$ and $3k$ where k is any whole number $\neq 1$ If 0 scored, award SC 1 mark for answer 25, 26, 28 or 25.4 to 25.5 or an answer that is a multiple of 32	May be done in stages $11+8+3$ implied by 22 May be implied by 3.1 to 3.2 or $\frac{35}{11}$ e.g. $22:16:6$, $33:24:9$, $44:32:12$ Accept $14k$ used in place of $11k$ and $3k$. Accept equivalents, e.g. $\frac{280}{11}$, $25\frac{5}{11}$
18	Correct position of town C drawn. 	2	Award 1 mark for one correct bearing drawn.	Tolerance $\pm 2^\circ$ for each bearing, use template. One correct bearing drawn can be implied by point C being inside one set of tolerance lines on the template (if no bearing line, there must be a point marked, not just the label. If more than one point, mark the point labelled C, if no label then apply the choice rule).
19	$p = t^2 - 3$	1		Accept any clear indication.
20	(\$) 2700	2	Award 1 mark for $[\frac{1}{3} \times] 135 \times k$ oe where $k = 10, 20, 30$ or 60 or better.	$\frac{1}{3} \times 135 \times k$ may be implied by $45 \times k$ 1 mark may be implied by 450, 900, 1350, 2700, 4050 $20 \times 135, 45 \times (10+20+30), [\frac{1}{3} \times] 8100$

Question	Answer	Marks	Part marks	Guidance																
21(a)	$R' = (6, 2)$	1																		
21(b)	$R' = (3, 7)$	1																		
22	6	1		Accept -6 or ± 6																
23	Correctly completed table of values, i.e. <table><tr><td>x</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>(8)</td><td>3</td><td>(0)</td><td>(-1)</td><td>(0)</td><td>3</td><td>8</td></tr></table> and A correct curve for the region $-3 \leq x \leq 3$ i.e. 	x	-3	-2	-1	0	1	2	3	y	(8)	3	(0)	(-1)	(0)	3	8	4	Award 2 marks for a correctly completed table of values or Award 1 mark for two correct values. and Award 2 marks for a correct curve or Award 1 mark for at least five points correctly plotted, FT <i>their</i> table.	Ignore curve outside of the region $-3 \leq x \leq 3$ for 1 or 2 marks. For this mark, curve can imply points. If more than 7 points plotted, firstly mark points on any curves/lines. In the absence of a curve, apply the choice rule.
x	-3	-2	-1	0	1	2	3													
y	(8)	3	(0)	(-1)	(0)	3	8													

Question	Answer	Marks	Part marks	Guidance
24	$(k =) 4$	2	Award 1 mark for $\frac{1}{2}kx(2x + 5)$ oe soi or $2x(2x + 5)$ or answer $k = 2$ or $k = 4x$	Accept expanded, e.g. $kx^2 + \frac{5kx}{2}$ Implied by, e.g. $[kx =] \frac{4x^2 + 10x}{2x + 5} \times 2$ $[kx =] \frac{8x^2 + 20x}{2x + 5}, [kx =] \frac{4x^2 + 10x}{x + 2.5}$ $k = 2x$ does not imply the factorising.

Question	Answer	Marks	Part marks	Guidance
25	51 (cents)	4	Award 3 marks for answer 5100 or Award 2 marks for correct method for perimeter, e.g. $4 \times \sqrt{\left(\frac{25.5}{2}\right)^2 + \left(\frac{34}{2}\right)^2}$ oe or better or Award 1 mark for $\left(\frac{25.5}{2}\right)^2 + \left(\frac{34}{2}\right)^2$ oe or better or $25.5^2 + 34^2$ oe or better and Award 1 mark for $\frac{k}{100} \times 60$ oe where k is any value they have previously written.	For all part marks, accept conversion to metres first. Or better, e.g. 85, 0.85 or $\frac{17}{20}$ oe, e.g. $2 \times \sqrt{25.5^2 + 34^2}$ For 1 mark, accept values below rounded or truncated to 3sf or better, e.g. $12.75^2 + 17^2$, $162.5625 + 289$ 451.5625, $\sqrt{12.75^2 + 17^2}$ $\sqrt{452}$, 21.25, $\frac{85}{4}$ e.g. $650.25 + 1156$, 1806.25, 42.5 May be done in stages. Do not accept being rounded up to an integer before multiplying by 60